Home walkthrough: kitchen4_lighter

10/25/2026, 3:30:02 PM · This patient (walker) · knees 50.0 cm, seated reach 60.0 cm

For the occupational therapist to review: a flag list, not a diagnosis. Measurements come from a 3D reconstruction of the home (Gaussian splat) scaled to metres; anything marked imagined is generated and never used for a number; **OT to verify** marks what the scan can't settle.

Expected toe catches per month

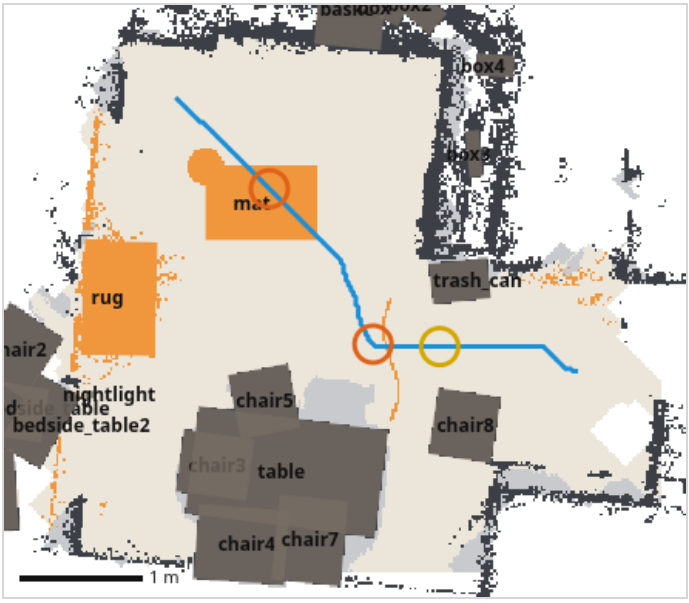
1.3

likely 1.2–1.3 · original layout 1.1 (**+0.2**)

200 simulated trips per route, gait "older_adult_walker". Toe-contact rate calibrated to published data (MTC 14.1 ± 8.3 mm lognormal shape; ~1 in 578 strides contact a 1 cm obstacle (Frontiers 2014)); darkness multiplier 1.5 (assumption). Range = measured height error + simulation noise.

Route	Trips/night	Contacts/month	Length
main route (night)	2	1.3 (1.2–1.3)	4.4 m

Layout and route





Map: grey = walls/furniture, orange = floor bumps (rugs, thresholds, cables), blue line = route, rings = hazards, blue furniture = moved, dashed = where it stood, light grey floor = scanned too thinly to trust (not walked).

Measured hazards on the route (main route)

What	Measurement	Along route	Contacts/month	
tight passage	3.5 cm each side of the walker	0.70 m	–	**OT to verify**
folded rug corner	4.2 cm high	1.79 m	1.2	**OT to verify**
cable	0.5 cm high **(estimated: classified, not measured)**	0.98 m	0.1	**OT to verify**

Transfers, reach and emergencies

Verdict	Check	Basis
fail	chair height 34 cm	measured from the splat; for knees at 50 cm the seat should be 47 cm–70 cm: low: hard to stand up
fail	chair2 height 33 cm	measured from the splat; for knees at 50 cm the seat should be 47 cm–70 cm: low: hard to stand up
fail	chair3 height 41 cm	measured from the splat; for knees at 50 cm the seat should be 47 cm–70 cm: low: hard to stand up
fail	chair4 height 41 cm	measured from the splat; for knees at 50 cm the seat should be 47 cm–70 cm: low: hard to stand up
fail	chair5 height 41 cm	measured from the splat; for knees at 50 cm the seat should be 47 cm–70 cm: low: hard to stand up
fail	chair7 height 41 cm	measured from the splat; for knees at 50 cm the seat should be 47 cm–70 cm: low: hard to stand up
fail	chair8 height 41 cm	measured from the splat; for knees at 50 cm the seat should be 47 cm–70 cm: low: hard to stand up
OT to verify	on the floor at the tight passage: no phone marked	farther than 3 m to crawl; consider a wearable alert
OT to verify	on the floor at the folded rug corner: no phone marked	farther than 3 m to crawl; consider a wearable alert
OT to verify	on the floor at the cable: no phone marked	farther than 3 m to crawl; consider a wearable alert

Changes in this layout

#	Change	Effort	Helpers	Cost	Product
none: the home as filmed					

Effort: moving an object grows with its mass (from the scan's measured size), distance carried and turn; Easy < 2, Moderate 2–3.5, Hard > 3.5. Helpers: two above 25 kg. Prices come from the product catalog; an unlisted item says "price needed" and is not added in.

Furniture weights (for move effort)

Object	Mass	1 m carry	Helpers	How we know
chair	5 kg	Easy	1	about 5 kg (from measured size)
chair2	5 kg	Easy	1	about 5 kg (from measured size)
bedside table	20 kg	Moderate	1	about 20 kg (from measured size)
bedside table2	19 kg	Moderate	1	about 19 kg (from measured size)
rug	1 kg	–	1	about 1 kg (from measured size)
chair3	5 kg	Easy	1	about 5 kg (from measured size)
chair4	7 kg	Easy	1	**measured size gave 7 kg, outside 4–7 kg for a chair: set to 7, check in person**
chair5	4 kg	Easy	1	about 4 kg (from measured size)
table	38 kg	Hard	2	about 38 kg (from measured size)
chair7	6 kg	Easy	1	about 6 kg (from measured size)
chair8	4 kg	Easy	1	about 4 kg (from measured size)
basket	3 kg	Easy	1	about 3 kg (from measured size)
box	7 kg	Easy	1	about 7 kg (from measured size)
box2	9 kg	Easy	1	about 9 kg (from measured size)
trash can	4 kg	Moderate	1	about 4 kg (from measured size)
box3	5 kg	Easy	1	about 5 kg (from measured size)
box4	6 kg	Easy	1	about 6 kg (from measured size)
mat	3 kg	–	1	about 3 kg (from measured size)
nightlight	0.5 kg	–	1	about 0.5 kg (guess by kind)

Recommended fixes, ranked

#	Fix	Toe catches/month saved	Passage gained	Cost	Time	Effort
1	**Remove the kitchen mat**	**−1.08** (95%)	–	free you can do it	about 2 min (estimate)	easy
2	**Tape the mat's edges down**	**−1.07** (94%)	–	$7–$8 DIY you can do it target.com	about 8 min (estimate)	easy

Each fix is simulated on its own from the layout above (200 trips per route); savings are not additive. Ranked by kitchen planner (B200 layout search + critic): each change alone on this layout, by toe catches/month saved (the same calculation as above), ties broken by the search objective (toe catches + simulated snags + brushes).

Planner options

Option	Toe catches/month	Price · install time · who	Effort	Changes from the scanned room
Easiest: most toe catches removed per unit of effort	0.07	free · about 2 min · you can do it	1.0	Remove the kitchen mat
Keep my mat: most toe catches removed per unit of effort without removing it	0.07	$7–$8 · about 8 min · you can do it	1.0	Tape the mat's edges down
Safest: fewest toe catches found	0.00	$13–$25 · about 10 min · you can do it	5.0	Move chair8 to (0.40, -0.20), turned 180 deg; Remove the kitchen mat; Tape the cord flat

Each option re-run on fresh simulated trips (200 per route), not the search's own draws.

Furniture the walker could push (simulated)

Object	Event	Route	Approach	Contact	P per touch/crossing (95% CI)	Per month
chair8	moves ≥ 2 cm	main route	9° · 0.50 m/s	28% of trips touch it	91% (83%–95%, n=95)	15.3
chair8	tips > 30°	main route	9° · 0.50 m/s	28% of trips touch it	0.0% (0.0%–3.9%, n=95)	0.0
mat	slides ≥ 2 cm	main route	45° · 0.50 m/s	every trip crosses it	66% (58%–74%, n=127)	39.7
mat	corner folds over	main route	45° · 0.50 m/s	every trip crosses it	2.4% (0.8%–6.7%, n=127)	1.4

SIMULATED (MuJoCo batch, physics/batch_mujoco.py): shapes from the scan; masses, frictions and corner stiffness sampled from assumed ranges, not measured; not yet checked against the real walker. Approach angle 0° = straight at the edge.

Methods and limits

- Measurements close to a limit are marked "borderline, OT to verify".
- Thin floor items (cables) are classified, not measured: their height is a typical value.
- Floor and walls under or behind furniture that was moved were never filmed; they are shown imagined (tinted) and not measured.
- Toe contacts are not falls; the numbers compare layouts for this person and route, they are not a fall probability.
- Fix prices and install times are US national ranges from published cost guides and retail listings (sources under each fix), not quotes: local labour rates, permits and what is inside the wall change them. Times marked "estimate" have no source.
- Furniture weights are estimated from the scanned size and a typical density per kind, not weighed; out-of-range values are marked for the OT.

Built from the run's log: 23 entries (computed 18 · guess 1 · simulated 3 · imagined 1).